Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/16/2025
PI Name:	Marr, David W.M. Ph.D.		
Project Title:	Dynamics of Complex Colloidal Molecules		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80401-1887	**Congressional District:**	7
Comments:	https://		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Ning Ph.D. (Colorado School of Mines)		
Grant/Contract No.:	80NSSC19K1725		
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Task Description:	Biopolymers such as polynucleotides and polypeptides are ubiquitous in nature. Although they have relatively simple backbones, by virtue of site-specific interactions between rather limited sets of subunits along the chains, they can fold into proteins and DNA with well-defined three-dimensional (3D) structures of exquisite complexity and functionality. In addition, biopolymers can also assemble into complex structures such as bundled microtubules, flagella, and cilia that undergo nonreciprocal motion in viscous liquid, which drive processes essential for life ranging from nutrient transport to pathogen clearance to cell migration. Scientists and engineers desire a similar capability to produce self-foldable and active functional materials; however, our knowledge of bottom-up assembly is limited by the difficulty in observing and characterizing the in situ processes at sufficiently small length and fast time scales. Unlike atoms and molecules, colloidal sizes can be conveniently tailored so that they are large enough to be probed individually. Colloidal particles are therefore excellent molecular analogues and their self- and directed-assembly has helped us understand fundamental		

	questions in material science including both equilibrium phase behavior and kinetic processes. However, the types of colloidal molecules studied to date have been limited to small sizes, simple symmetries, and rigid structures. We remain far from accessing the full diversity of fundamentally and technologically relevant structures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In-situ manipulation of interactions and dynamic pathways, based on the use of external fields, could lead to crystalline structures beyond those seen in Nature. Our studies move significantly beyond simple hard spheres and towards experimental models appropriate for studying fundamental questions associated with complex symmetries. Results from this study will lead to improved control of non-covalent assembly of molecules, efficient tailoring of lattice symmetries, and the scalable processing of nano-structured materials. The development of new colloidal molecules and their associated assembled structures may have considerable technological impact as well. Anisotropic interactions,. both at the particle level and via applied fields, have been shown generally to lead to arrays with reduced symmetry and enhanced directionality. Such structures interact with a broad range of electromagnetic radiation in unique ways and can. They can exhibit collective photonic, plasmonic, mechanical, electronic, or magnetic properties that are not manifested at the level of single particles. As a result, they have significant potential as next-generation functional materials. For example, dielectric arrays with diamond-like or quasicrystalline lattices have been proposed as ideal photonic crystals that can manipulate light propagation in three dimensions, forming the basis for next-generation all-optical communication technologies. In addition, colloidal networks are common in a range of pharmaceutical and consumer production applications because of their ability to impart a yield stress that can stabilize active components uniformly throughout the formulation. Developing methods to produce colloidal networks with defined coordination, connectivity, and topology would expand the use and performance of colloidal stabilizing networks in these materials. Here we use a combination of applied external fields to control the assembly of colloidal particles into new structures and building blocks. Specifically, a combination of an in-plane rotating magnetic field and an out-of-plane electric field has yielded quasicrystals, a morphology that cannot be obtained by either of the fields alone. Only recently discovered, quasicrystals have enjoyed significant recent interest because of their unusual formation mechanisms and high rotational symmetry, which may lead to structures with unique physical and photonic properties. In fact, the quasicrystalline structures that have been made possible by this research have not been previously reversibly and spontaneously fabricated at such large length scales.
Task Progress:	Our progress during this period includes: 1. Identifying the magnetic/electric field combinations necessary for the formation of colloidal quasicrystals. 2. Developing characterization methods for quantifying quasicrystalline morphology. 3. Characterizing quasicrystalline formation kinetics.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Gao Y, Sprinkle B, Marr DWM, Wu N. "Direct observation of colloidal quasicrystallization." Nat. Phys. 2025 Mar 31:1-8. https://doi.org/10.1038/s41567-025-02859-z , Mar-2025
Articles in Peer-reviewed Journals	Yang T, Sprinkle B, Guo Y, Qian J, Hua D, Donev A, Marr DW, Wu N. "Reconfigurable microbots folded from simple colloidal chains." Proc Natl Acad Sci USA. 2020 Aug 4;117(31):18186-93. https://doi.org/10.1073/pnas.2007255117 ; PMID: 32680965; PMCID: PMC7414297 , Aug-2020